DIGITAL DASHBOARD

Developer and Administrator Guide

Unified architecture, analytical integration, deployment, and maintenance

Documentation revision	June 2026
Language	English
Status	Aligned with the supplied current application source

Purpose. This document describes the current workflow and implementation represented by the supplied project files. It replaces older instructions that referred to previous navigation, storage, or output structures.

Contents

1. Current architecture

- Repository map
- Module split

2. Startup and dependencies

- Source order
- Packages

3. Shared state and file import

- State invalidation
- Current format limitation

4. Unified filter behavior

- CUSUM versus Geospatial
- Invalidation

5. Analytical execution

- CUSUM
- Geospatial

6. Display and output system

- Display settings
- Dynamic tabs
- Downloads

7. Configuration and spatial data

- config.json
- GeoPackages

8. Documentation and Help links

- Current links
- Case sensitivity

9. Testing and release checks

- Minimum matrix
- Regression risks

10. Deployment and maintenance

- Packaging
- Troubleshooting

1. Current architecture

```
DigitalDashboard/
|-- app.R
|-- global.R
|-- config.json
|-- raw_data/                    # Required GeoPackages at deployment
|-- R/
|   |-- shared/{app_state,data_import,file_manager,ui_components}.R
|   |-- cusum/{functions_*.R, mod_cusum.R}
|   |-- geospatial/{functions_*.R, mod_geospatial.R}
|   `-- modules/
|       |-- mod_app.R
|       |-- mod_unified_dashboard.R
|       `-- dashboard/
|           |-- dashboard_sidebar_ui.R
|           |-- dashboard_main_ui.R
|           |-- dashboard_display_ui.R
|           |-- dashboard_tables_ui.R
|           |-- dashboard_file_server.R
|           |-- dashboard_filter_server.R
|           |-- dashboard_cusum_server.R
|           |-- dashboard_geo_server.R
|           `-- dashboard_output_server.R
`-- www/{config.css, docs/}
```

appUI() creates hidden Home, Dashboard, Display, and Help tab panels controlled by a custom navigation bar. unifiedDashboardServer() coordinates file, filter, CUSUM, geospatial, and output server components through a shared reactive state object.

2. Startup and dependencies

global.R sources shared helpers, analytical engines, dashboard pieces, and the two coordinators in a deliberate order. It then disables s2, reads config.json, and currently fixes active_country to Argentina, target_var to prevalence_100k, and is_offline to FALSE.

Area	Packages
UI	shiny, shinyjs, DT, htmltools, jsonlite
Data	dplyr, tidyr, purrr, rlang, readr, lubridate, ISOweek, stringr
Plots	ggplot2
Spatial	sf, spdep, leaflet
Utilities	here

```
shiny::runApp()
```

Current country behavior. Unlike the standalone applications, the supplied Dashboard source does not expose a country selector and assigns active_country <- "Argentina" in global.R. Configuring other countries requires code and configuration review.

3. Shared state and file import

create_app_state() holds uploaded-file metadata, active IDs, combined raw data, file names, errors, CUSUM prepared/results, and geospatial results. refresh_active_dashboard_data() rebuilds raw_data from active files and invalidates both analytical result sets.

The file input accepts .csv, .txt, .tsv, and .xlsx, but read_api_pop_files() currently delegates to a CSV delimiter reader. This makes comma- and semicolon-delimited text the supported implementation. Either restrict the UI accept list or add explicit TSV/XLSX readers before advertising those formats.

Standardisation	Current behavior
Column names	Lower-case and trim.
Case aliases	n_cases, cases, case_count, count, n.
Population aliases	population, n_population, pop, denominator.
Numeric conversion	year, week, n_cases, population.
Geographic trimming	country, level1-level4.
Code padding	Uses code_length from config mappings.

Validation difference. Dashboard validate_api_pop_data() requires country, year, week, n_cases, and a level column. at least one level colu Population is required for manual CUSUM mu0 and Geospatial execution.

4. Unified filter behavior

dashboard_filter_server() populates Geographic Level, Locations, Year, and the Map Time Window. Locations are selected for all units by default.

Input	CUSUM server	Geospatial server
global_level	Used to create CUSUM analysis_unit_id.	Used to select the configured map layer.
target_locations	Required and used to filter CUSUM preparation.	Not referenced by dashboard_geo_server().
detection_period	CUSUM window and baseline calculation.	Not used.
selected_year	Not used by CUSUM execution.	Filters map data.
week_range / time_mode	Not used by CUSUM execution.	Filters single week or inclusive range.

Changing global_level clears all results. Changing target_locations clears only CUSUM prepared/results. User notifications explicitly instruct a rerun.

Documentation contract. Until the code changes, user material must state that Locations affects CUSUM only and that Geospatial includes all units at the selected level.

5. Analytical execution

CUSUM

prepared_cusum_data() calls process_target_data() with overlap_method = "sum" and the selected locations. Automatic expected counts are resolved per unit, k is derived from RR, and run_cusum_all_units() executes with reset = TRUE.

Geospatial

dashboard_geo_server() validates population, normalises single/range weeks, calls prepare_geospatial_dataset(), loads the GeoPackage layer, pads spatial IDs, joins, and computes LISA. Range mode enforces at least a two-week interval when the year contains multiple weeks.

The Dashboard copies analytical helper functions into its own R/cusum and R/geospatial folders. Changes made in standalone RAP repositories do not automatically propagate; maintain explicit synchronization or shared-package versioning.

6. Display and output system

dashboard_display_ui.R writes namespaced inputs that are consumed by dashboard_output_server.R even though Display is an app-level page. The output server defaults to CUSUM overview plus both maps, two columns, and tables inside Output.

Setting	Implementation
visible_outputs	Controls output-card creation.
layout_columns	One- or two-column Bootstrap layout.
table_display_mode	Tables inside Output or a dynamic Output Table tab.
display_font_size	Applies a body CSS class and ggplot theme sizes.
display_decimals	0-6 decimal formatting for DT tables.
reactive_display	Re-runs eligible CUSUM or Geospatial server functions after an initial result.

dashboard_internal_tabs is assembled dynamically from Output, optional Output Table, and Prepared Data. Download buttons are rendered only when data exist. CSV downloads use underlying data; map screenshots depend on optional packages.

7. Configuration and spatial data

The supplied config.json contains country mapping blocks but lacks several top-level defaults used by the standalone Geospatial app. The Dashboard uses hard-coded active_country, target_var, and is_offline values from global.R.

Each selectable mapped level requires csv_column, layer_name, spatial_id, label_col or name_col, and usually code_length. Place the corresponding GeoPackage under raw_data using the country file_name.

Consolidation opportunity. To reduce drift, move active country, target variable, rate multiplier, defaults, and pre-flight logic into shared configuration helpers consistent with the standalone apps.

8. Documentation and Help links

The Help page links to docs/cusum/HELP_en.pdf, docs/cusum/HELP_es.pdf, docs/geospatial/HELP_en.pdf, and docs/geospatial/HELP_es.pdf. Linux deployments are case sensitive. The updated documentation package includes these exact lowercase aliases.

The Help page currently describes Geospatial location filters as a subset. That wording does not match dashboard_geo_server() and should be corrected in R/shared/ui_components.R during the next code update.

Recommended primary manuals. Link the Help page to the dedicated Digital Dashboard user manual first, with CUSUM and Geospatial concept notes as supplemental documents.

9. Testing and release checks

- Upload one and multiple CSV files; activate/deactivate/remove files.
- Run CUSUM with automatic and manual expected frequency.
- Change level and selected locations and verify invalidation.
- Run single-week and week-range maps.
- Exercise both table placements, one/two columns, all font sizes, decimals, and reactive display.
- Verify all downloads and Help links on Windows and Linux-like case-sensitive paths.
- Compare copied analytical helpers with the latest standalone repositories.

Test gap. The supplied Dashboard archive does not include a tests/ directory. Add automated tests for data import, state invalidation, filter semantics, and analytical helper synchronization.

10. Deployment and maintenance

- Run from the project root.
- Package R/, www/, config.json, and required raw_data GeoPackages.
- Verify GDAL/PROJ and R spatial packages.
- Use approved hosting for restricted data.
- Limit uploads and avoid logging row-level data.
- Keep documentation filenames aligned with Help links.

Symptom	Likely area
File picker accepts but import fails	Mismatch between accept list and CSV-only reader.
CUSUM changes but maps do not	Different filter semantics or reactive_display eligibility.
Locations do not filter maps	Current dashboard_geo_server design.
Output Table missing	table_display_mode is output or no results exist.
Prepared tab missing	No prepared/result objects are available.
Help PDF 404 on Linux	Filename case mismatch.
Standalone fix missing in Dashboard	Copied helper files were not synchronized.